

Math 548
Portfolio Assignment 6

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Problem 1 (S25, P2). Let S_n denote the symmetric group on n letters.

- (a) Is the element $(1234)(25346)(153247) \in S_7$ even or odd? Indicate your reasoning.
- (b) Find the order of $(134)(243)(134) \in S_4$. Show all work.
- (c) Write $(1523)(2134)(1523)^{-1} \in S_5$ in disjoint cycle form. Show all work.

Proof. (a) Note that:

$$\begin{aligned}(1234)(25346)(153247) &= (1472635) \\ &= (15)(13)(16)(12)(17)(14).\end{aligned}$$

Since $(1234)(25346)(153247)$ can be written as six transpositions, $(1234)(25346)(153247)$ is even.

(b) The order of $(134)(243)(134)$ is the least common multiple of the disjoint cycle lengths. Hence the order of $(134)(243)(134)$ is 3.

(c) We have:

$$\begin{aligned}(1523)(2134)(1523)^{-1} &= (1523)(2134)(3251) \\ &= (1435).\end{aligned}$$

□

Problem 2 (S22, P2). The dihedral group D_4 is the group of 8 rigid motions of a square. Prove that D_4 is not the internal direct product of two of its proper subgroups.

Proof. Suppose towards contradiction we can write $D_8 = H \times K$, where $H \times K$ are proper subgroups of D_8 . Without loss of generality, suppose $|H| = 2$ and $|K| = 4$. Since $H \times K$ is an internal product, for all $h \in H$ and all $k \in K$ we have $hk = kh$. Moreover, since $|H| = 2$, we know that H is abelian. Likewise, there up to isomorphism two groups of order 4, the groups $\mathbb{Z}/4\mathbb{Z}$ and $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$. Both of these groups are abelian, hence K is abelian. From this, we can conclude $H \times K$ is abelian. But this is a contradiction, as D_8 is not abelian. □

Problem 3 (W20, P3). Let G be a group. For each $a \in G$, let γ_a denote the automorphism of G defined by $\gamma_a(b) = aba^{-1}$ for all $b \in G$. The set $\text{Inn}(G) = \{\gamma_a \mid a \in G\}$ is a subgroup of the automorphism group of G , called the subgroup of **inner automorphisms**. Prove that $\text{Inn}(G)$ is isomorphic to $G/Z(G)$, where $Z(G)$ is the center of G .

Proof. Define $\varphi : G \rightarrow \text{Inn}(G)$ by $g \mapsto \gamma_g$. Then φ is a homomorphism since:

$$\begin{aligned}\varphi(g_1 g_2)(x) &= \gamma_{g_1 g_2}(x) \\ &= g_1 g_2 x g_2^{-1} g_1^{-1} \\ &= g_1(\gamma_{g_2}(x))g_1^{-1} \\ &= g_1(\varphi(g_2)(x))g_1^{-1} \\ &= \varphi(g_1)(\varphi(g_2)(x)) \\ &= (\varphi(g_1) \circ \varphi(g_2))(x).\end{aligned}$$

Now let $\psi \in \text{Inn}(G)$. Then $\psi(x) = gxg^{-1}$ for some $g \in G$. Hence φ is surjective, as $\varphi(g)(x) = \gamma_g(x) = gxg^{-1} = \psi(x)$. Finally, observe that:

$$\begin{aligned}\ker\{g \in G \mid \varphi(g)(x) = \text{id}(x), x \in G\} \\ &= \{g \in G \mid \gamma_g(x) = \text{id}(x), x \in G\} \\ &= \{g \in G \mid gxg^{-1} = x, x \in G\} \\ &= \{g \in G \mid gx = xg, x \in G\} \\ &= Z(G).\end{aligned}$$

By the First Isomorphism Theorem, $G/Z(G) \cong \text{Inn}(G)$. □

Problem 4 (W19, P3). Let G be a group and suppose $\text{Aut}(G)$ is trivial.

- (a) Show that G is abelian.
- (b) Show that for any abelian group H , the **inversion map** $\phi(h) = h^{-1}$ is an automorphism.
- (c) Use parts (a) and (b) above to show that g^2 is the identity element for every $g \in G$.

Proof. (a) Define $\gamma_g : G \rightarrow G$ by $\gamma_g(x) = gxg^{-1}$. This is an automorphism. In particular, since $\text{Aut}(G)$ is trivial, it must be the case that $\gamma_g = \text{id}$. Hence for all $x, g \in G$, we have $\gamma_g(x) = \text{id}(x)$; i.e., $gxg^{-1} = x$. Thus $gx = xg$, hence G is abelian.

(b) For any $h_1, h_2 \in H$, we have:

$$\begin{aligned}\phi(h_1 h_2) &= (h_1 h_2)^{-1} \\ &= h_2^{-1} h_1^{-1} \\ &= h_1^{-1} h_2^{-1} \\ &= \phi(h_1) \phi(h_2).\end{aligned}$$

Hence ϕ is a homomorphism. Moreover, note that ϕ is its own inverse since $\phi(\phi(h)) = (h^{-1})^{-1} = h$. Hence ϕ is a bijection. Therefore ϕ is an automorphism.

(c) By part (b) we have $\phi \in \text{Aut}(G)$. Hence $\phi(g) = \text{id}(g)$ for all $g \in G$. This means $g^{-1} = g$; i.e., $e_G = g^2$. □